

The cycle type of two rows of a random Latin square: an exact ladder identity and a conditional proof of the weak Cavenagh–Greenhill–Wanless conjecture

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Abstract

Let L be a uniformly random Latin square of order n and σ the permutation of columns carrying its first row to its second. Cavenagh, Greenhill and Wanless conjectured that the cycle type of σ is asymptotically that of a uniform derangement, in total variation. We prove three exact identities — the marked-pair identity, which expresses the ratio of completion counts of adjacent cycle types as the odds of one bit (“rows 1, 2 lie in a common column cycle of a marked column pair”) in a single uniform square; a reduction theorem, by which the conjecture follows from that bit being fair to $o(1/\log n)$; and the ladder identity, by which the bit is exactly fair except on the squares in which no “ladder pair” of rows is flippable — and an orbit method by which the probability of the exceptional set is bounded by a product of conditionally independent factors. With coordinates chosen along the row permutation of the ladder pair from the marked columns, each coordinate is placed where it toggles, and the conjecture is thereby reduced to a hypothesis about short column cycles at the ladder rows (Hypothesis 5.6): that with probability $1 - o(1/\log n)$ some $C \log \log n$ of the $n/\log^2 n$ ladder pairs carry a short clean non-colliding cycle at one of $O(\log^2 n)$ prescribed column pairs, and likewise for the offset diagonals of short-arc instances. We state it precisely, test its ingredients numerically up to $n = 100$, and locate it within the known switching technology. Along the way we observe that the published proof of the upper bound in Lemma 3.12 of Cavenagh–Greenhill–Wanless has a gap for every split other than $(2, 2)$; our results do not depend on that bound, and we re-derive from the surviving direction the tail estimate on the number of cycles that the reduction theorem needs.

1 Introduction

Let P_n be the law of the cycle type of $\sigma = \sigma_{1,2}(L)$ for a uniform Latin square L of order n , and Q_n the law of the cycle type of a uniform derangement of $[n]$. Cavenagh, Greenhill and Wanless [1, Conj. 6.1] conjectured:

Conjecture 1.1. $d_{\text{TV}}(P_n, Q_n) = o(1)$ as $n \rightarrow \infty$.

For $\lambda \vdash n$ with all parts ≥ 2 let $C_n(\lambda)$ be the number of completions of a fixed $2 \times n$ Latin rectangle of cycle type λ to a Latin square, divided by $(n-2)!$ (the count does not depend on the rectangle chosen, since rectangles of the same type are isotopic by a row-preserving isotopy), and let $\gamma(\lambda)$ be the number of permutations of type λ . Then $P_n(\lambda) \propto \gamma(\lambda)C_n(\lambda)$ and $Q_n(\lambda) \propto \gamma(\lambda)$, so the conjecture says that C_n is nearly constant on the types that carry most of the mass. CGW proved $C_n(\lambda)/C_n(\mu) \in [\frac{1}{2}, \frac{3}{2}]$ for adjacent types (one part of λ split into two parts of μ ,

both ≥ 2), and used it to show that the two laws agree within a factor $n^{3/2}$ on every type with at most $\frac{6}{5} \log n$ parts; the upper bound $\frac{3}{2}$, and with it the upper half of the latter statement, is affected by the gap described in Remark 3.4 and [2].

Results. Throughout, a *mark* is a column pair $P = \{j, j'\}$ with $j' = \sigma^\alpha(j)$ on an m -cycle of σ , $m = \alpha + \beta$, $\alpha, \beta \geq 2$; $X = X(n, \lambda; \alpha, \beta)$ is the set of marked squares of type λ ; the *frame* of P is the permutation of rows $\pi(r) = r'$ with $L(r', j') = L(r, j)$; and P is a *B-pair* if rows 1, 2 lie in one cycle of π , an *A-pair* otherwise. Our results are:

- (1) *Marked-pair identity* (Theorem 2.4): $2C_n(\lambda)/C_n(\mu) - 1 = \mathbb{P}_X[B]/\mathbb{P}_X[A]$ exactly, for every $n, \lambda, \alpha, \beta$.
- (2) *Reduction* (Theorem 3.3): if $|\mathbb{P}_X[B] - \frac{1}{2}| \leq \delta(n)/4$ for all adjacent pairs with $\alpha \neq \beta$, then $d_{TV}(P_n, Q_n) = O(\delta \log n + n^{-1+o(1)})$. The tail estimate it needs on the number of cycles of σ is proved from the trivial half of (1) alone (Proposition 3.2).
- (3) *Ladder identity* (Theorem 4.2): with the ladder pairs $(x_k, y_k) = (\pi^{-k}(1), \pi^{-k}(2))$, $1 \leq k < K_0$, and “flippable” meaning that j, j' lie in different cycles of the column permutation of rows x_k, y_k ,

$$\mathbb{P}_X[B] - \mathbb{P}_X[A] = \mathbb{P}_X[B, \text{Flip} = \emptyset] - \mathbb{P}_X[A, \text{Flip} = \emptyset], \quad |\mathbb{P}_X[B] - \frac{1}{2}| \leq \frac{1}{2} \mathbb{P}_X[\text{Flip} = \emptyset].$$

So Conjecture 1.1 follows from (L): $\mathbb{P}_X[\text{Flip} = \emptyset] = o(1/\log n)$.

- (4) *Orbit method* (Propositions 5.3 and 5.5): the flippabilities of a family of ladder pairs are conditionally independent given the orbit of a group of commuting column-cycle turns. With a fixed matching of columns as coordinates this gives $\mathbb{P}_X[\text{Flip}_I = \emptyset] \leq \mathbb{E}_X \exp(-\frac{1}{2} \sum_{k \in I} \bar{y}_k)$ with an explicit “orbit weight” $\bar{y}_k$ that needs a genericity input; with the *adaptive* coordinates $(\rho_k^t(p), \rho_k^t(p'))$ — the t -th columns along the row permutation ρ_k of the k -th ladder pair from the marked columns — every coordinate is separated by the mark, so its turn toggles whenever it is legal and clean, and $\mathbb{P}_X[\text{Flip}_I = \emptyset] \leq \mathbb{E}_X[2^{-|U|}]$, where U is the set of ladder pairs having a short clean candidate cycle (subject to a collision condition). Hence (L), and the conjecture, follow from Hypothesis 5.6 (Proposition 5.7).

Hypothesis 5.6 says that, with probability $1 - o(1/\log n)$, at least $C \log \log n$ of the first $n/\log^2 n$ ladder pairs carry a short clean column cycle through their x -row at one of $O(\log^2 n)$ prescribed column pairs — a lower-tail statement about rare local configurations (each determined by $O(\log^2 n)$ cells) at rows selected by the square, under the conditioning on the type of rows 1, 2. Its heuristic expectation is $\asymp \log^2 n$ such pairs. No genericity constant enters; the fixed-matching version needed one (a probability $\geq c$ that a generic column pair is interleaved with the mark in ρ_k , measured at 0.19 and not provable by the switchings available inside X , §6). The data give $\mathbb{P}_X[B] = 0.49\text{--}0.50$ (± 0.01) at $n = 30, 50, 100$.

A gap in CGW’s Lemma 3.12. Case 3 of the splitting procedure in [1] (cross-switch at $\{\omega j, \omega j'\}$, then backflip at $\{j, j'\}$) does not behave as claimed: the cross-switch reverses the ω -arc from ωj to $\omega j'$ (or the complementary arc, depending on the smallest corner symbol), which contains j' (resp. j), so $\{j, j'\}$ ends at distance 2. For $\min(\alpha, \beta) \geq 3$ the backflip then lands in the wrong type; for $\min(\alpha, \beta) = 2 < \max(\alpha, \beta)$ the type is right but the edge is not an edge of the joining multigraph, so the identity $G_S = G_J$ on which the double count rests

fails (already at $n = 5$, where we computed both multigraphs exactly). We give the details in a separate note [2]; here we only record (Remark 3.4) which published statements depend on it. Nothing in the present paper does: the lower bound $\frac{1}{2}$ of CGW’s lemma is the trivial half of Theorem 2.4, and the upper bound $\frac{3}{2}$ is, by the same theorem, $\mathbb{P}_X[B] \leq \frac{2}{3}$ — a weak form of what we are after. By Theorem 4.2 it would follow from $\mathbb{P}_X[\text{Flip} = \emptyset] \leq \frac{1}{3}$.

Conventions. $L(r, c)$ is the entry in row r , column c . For rows x, y , $\rho_{x,y}$ is the permutation of columns with $L(y, \rho_{x,y}(q)) = L(x, q)$, so that $\sigma = \rho_{1,2}$ up to inversion; a *row cycle* of (x, y) is a cycle of $\rho_{x,y}$. For columns q, c , $\delta_{q,c}$ is the permutation of rows with $L(\delta(r), c) = L(r, q)$, and its cycles are the *column cycles* of (q, c) . A *trade* of rows x, y along a row cycle swaps their entries on the columns of that cycle; a *turn* of a column cycle swaps the entries of columns q, c in its rows. Both are Latin trades (the result is again a Latin square). For $\lambda \vdash n$, $S(\lambda)$ is the set of Latin squares of order n with first row the identity whose rows 1, 2 have cycle type λ , so $|S(\lambda)| = \gamma(\lambda)(n-2)!C_n(\lambda)$; we also write $D_\lambda = \gamma(\lambda)$ for the number of derangements of type λ . A column pair $\{j, j'\}$ is *admissible* if $j' \notin \{j, \sigma(j), \sigma^{-1}(j)\}$ (CGW’s condition for the pair operations; every marked pair is admissible since $\alpha, \beta \geq 2$). Following CGW, the *flip* at an A -pair whose columns lie in different σ -cycles, and the *backflip* (or *unflip*) at an A -pair whose columns lie in one σ -cycle, is the turn of the column cycle of the pair through row 2 (it joins, resp. splits, the σ -cycles involved); the *switch* at a pair whose columns lie in different σ -cycles is the trade of rows 1, 2 along the row cycle through one of the two columns, selected by CGW’s rule (the cycle whose columns carry the smaller minimum symbol). We call Conjecture 1.1, which asks only for convergence in total variation, the *weak* conjecture; the stronger statements suggested by Cameron (“tends very rapidly”) and by the data (pointwise ratios $1 + o(1)$) are not addressed here. Jacobson–Matthews sampling [3] is used for all numerical data; the scripts and logs are in the project repository.

2 The marked-pair identity

Throughout, squares are normalised to have first row id; CGW’s column map ω , defined by $L(1, \omega(j)) = L(2, j)$, is then $\omega(j) = L(2, j)$, and we take $\sigma = \omega$ (only the cycle type matters, and it is the same for ω and ω^{-1}). For the proofs it is convenient to work with *unnormalised* squares (equivalently, symbol-relabelling classes): every instance count below then acquires a common factor $n!$, which cancels from every ratio, and the maps we use (flip, switch) act on unnormalised squares without any renormalisation step. Statuses, cycle types, and column positions are relabelling-invariant, so we freely state results in the normalised picture.

Definition 2.1 (marked instances). Let $\lambda \vdash n$ contain a part $m = \alpha + \beta$ ($\alpha, \beta \geq 2$), and let μ be λ with one part m split into α, β . The *splitting instance space* is

$$X = X(n, \lambda; \alpha, \beta) = \{(L', P) : L' \in S(\lambda), P = \{j, \sigma^\alpha(j)\} \text{ with } j \text{ in an } m\text{-cycle of } \sigma(L')\},$$

unordered pairs counted once. Thus every L' carries exactly $\lambda_m m$ marked pairs when $\alpha \neq \beta$ and $\lambda_m \alpha$ when $\alpha = \beta$. Since $\alpha, \beta \geq 2$ every marked pair is admissible, and $X = X_A \sqcup X_B$ according to the A/B status of P in L' . The *joining instance space* is

$$J = J(n, \mu; \alpha, \beta) = \{(L, P) : L \in S(\mu), P = \{j, j'\}, j \text{ in an } \alpha\text{-cycle, } j' \text{ in a different } \beta\text{-cycle}\},$$

again with $J = J_A \sqcup J_B$ by the status of P in L .

The counting arithmetic is immediate: with D_ν the number of derangements of cycle type ν and $N(\nu) = C_n(\nu)(n-2)!$,

$$\frac{2|X|}{|J|} = \frac{2 D_\lambda N(\lambda) \lambda_m m}{D_\mu N(\mu) \mu_\alpha \mu_\beta \alpha \beta} = \frac{2 C_n(\lambda)}{C_n(\mu)} \quad (\alpha \neq \beta), \quad (1)$$

using $D_\lambda/D_\mu = \alpha \mu_\alpha \beta \mu_\beta / (m \lambda_m)$; for $\alpha = \beta$ one has $D_\lambda/D_\mu = \alpha^2 \mu_\alpha (\mu_\alpha - 1) / (m \lambda_m)$, $|J| = |S(\mu)| \binom{\mu_\alpha}{2} \alpha^2$ and $|X| = |S(\lambda)| \lambda_m \alpha$, and (1) holds verbatim. The content of the identity is therefore the following count.

Lemma 2.2 (cycle trades preserve pair status). *Let $P = \{j, j'\}$ be an admissible column pair of a Latin square L and let C be one column cycle of P . Trading C (swapping the entries of columns j, j' in the rows of C) inverts the pair permutation π_P on C and fixes it elsewhere; the partition of rows into column cycles of P is unchanged. In particular the A/B status of P itself is invariant under the CGW flip at P , which trades the full column cycle through row 2.*

Proof. Write $\pi = \pi_P$, so $\pi(r)$ is the row of column j' holding the symbol $L(r, j)$; C is π -closed and the trade swaps $L(r, j) \leftrightarrow L(r, j')$ for $r \in C$. Fix $r \in C$ and let π' be the new pair permutation. The new $L(r, j)$ is the old $L(r, j')$; the row of the new column j' holding that symbol is found by solving (new $L(s, j')$) = (old $L(r, j')$): for $s \in C$ the new $L(s, j')$ is the old $L(s, j)$, and old $L(s, j) = \text{old } L(r, j')$ exactly when $\pi(s) = r$. Hence $\pi' = \pi^{-1}$ on C ; off C nothing changes. A cycle inverted in place has the same support, so the partition into cycles is unchanged. $\square$

Lemma 2.3 (switch toggle). *Let $(L, P) \in J$ and let Q be the row cycle through one column of P (the two row cycles are distinct), selected by CGW's min-symbol rule. Trading Q (the CGW switch) changes π_P by precomposition with the transposition of rows 1, 2: the cycles of π_P through rows 1, 2 are split if rows 1, 2 are in one cycle, and joined if in two. Hence the switch toggles the status of P , preserves $S(\mu)$ and the instance data (it inverts one σ -cycle in place), and is an involution: the trade preserves the traded cycle's symbol set (the two rows carry the same symbols on its columns), so the min-symbol rule re-selects the same cycle. Consequently $|J_A| = |J_B|$, exactly.*

Proof. Say Q is the row cycle through j ; its columns include j but not j' . The trade swaps the row-1 and row-2 entries in the columns of Q ; for π_P only column j matters, where the entries of rows 1, 2 are exchanged. Thus $\pi'_P = \pi_P \circ \tau_{12}$, and multiplying by a transposition splits or joins the cycles through its two points. Rows 1, 2 lie in one cycle of π_P exactly when P is a B -pair. The trade swaps the two rows' entries along the columns of Q , which inverts the corresponding σ -cycle in place: the cycle type and the column sets of all row cycles are preserved, so $(L, P) \mapsto (\text{sw}(L), P)$ maps J to J . $\square$

Theorem 2.4 (marked-pair identity). *For every n , λ , and split $m = \alpha + \beta$ with $\alpha, \beta \geq 2$ (including $\alpha = \beta$),*

$$|J_A| = |J_B| = |X_A|, \quad \text{hence} \quad \boxed{\frac{C_n(\lambda)}{C_n(\mu)} = \frac{|X|}{2|X_A|}, \quad 2 \frac{C_n(\lambda)}{C_n(\mu)} - 1 = \frac{|X_B|}{|X_A|}.}$$

In particular $C_n(\lambda) > \frac{1}{2} C_n(\mu)$ unconditionally, and Hypothesis 3.1 (EQ(δ)) is equivalent to

$$\left| \mathbb{P}(P \text{ is a } B\text{-pair}) - \frac{1}{2} \right| \leq \frac{\delta}{4}(1 + o(1)),$$

for (L', P) uniform on X : the law of a single bit — whether rows 1, 2 lie in one column cycle of the marked pair $\{j, \sigma^\alpha(j)\}$ — of a single uniform square of $S(\lambda)$.

Proof. By (1) it suffices to prove $|J| = 2|X_A|$. The flip at an A -pair P whose columns lie in one row cycle at σ -distances (α, β) splits that cycle into an α - and a β -cycle and maps $S(\lambda) \rightarrow S(\mu)$; by Lemma 2.2 the pair P is still an A -pair afterwards, so $x \mapsto \text{unflip}(x)$ maps X_A into J_A . The flip at a fixed pair is an involution (the traded cycle again contains row 2, with the same row support), so $\text{join} = \text{flip}$ inverts it: $X_A \leftrightarrow J_A$ is a bijection. Composing with the switch of Lemma 2.3 gives a bijection $X_A \leftrightarrow J_B$, namely $x \mapsto \text{sw}(\text{unflip}(x))$ with inverse $(L, P) \mapsto \text{flip}(\text{sw}(L))$ — the map “switch-then-flip”, which is exactly the CGW joining step at a B -pair. Hence $|J| = |J_A| + |J_B| = 2|X_A|$. $\square$

3 The reduction theorem

Hypothesis 3.1 (B-pair equidistribution, $\text{EQ}(\delta)$). *For every $\lambda \vdash n$ with a part $m = \alpha + \beta$, $\alpha \neq \beta$, $\alpha, \beta \geq 2$, and μ the split type, $|\bar{q}(\mu \rightarrow \lambda) - 1| \leq \delta(n)$, where $\bar{q}(\mu \rightarrow \lambda) := 2C_n(\lambda)/C_n(\mu) - 1 = \mathbb{P}_X[B]/\mathbb{P}_X[A]$ by Theorem 2.4.*

By Theorem 2.4, $\text{EQ}(\delta)$ is the statement $|\mathbb{P}_X[B] - \frac{1}{2}| \leq \frac{\delta}{4}(1 + o(1))$ for all adjacent pairs with $\alpha \neq \beta$. The proof below needs a tail bound on the number κ of cycles of σ ; we first prove it from the trivial half of Theorem 2.4 alone, so that nothing depends on the upper bound of [1, Lemma 3.12] (Remark 3.4).

Proposition 3.2 (the number of cycles, from the surviving direction only). *Let L be a uniform Latin square of order n , σ the column permutation of rows 1, 2, N_ℓ its number of ℓ -cycles and $\kappa = \sum_\ell N_\ell$. Then, using only the identity $|J| = 2|X_A|$ of Theorem 2.4 and $|X_A| \leq |X|$:*

(i) $\mathbb{E}[\ell N_\ell] \leq 2n/(n - \ell)$ for every $\ell < n$; in particular $\mathbb{E}[N_\ell] \leq 4/\ell$ for $\ell \leq n/2$.

(ii) For every $k \geq 1$, with $(\kappa)_k = \kappa(\kappa - 1) \cdots (\kappa - k + 1)$,

$$\mathbb{E}[(\kappa)_k] \leq 2^{k-1} \sum_{\ell_1, \dots, \ell_k \geq 2} \frac{\mathbb{E}[m N_m]}{\ell_1 \cdots \ell_k} \Big|_{m=\ell_1+\dots+\ell_k} \leq 2^{k-1} \left(4 + \frac{k^2}{\log n}\right) (\log n)^k.$$

(iii) For $A \geq 3$, $\mathbb{P}[\kappa \geq A \log n] \leq (7 + \log n) n^{-\log((A-1)/2)}$, which is $O(n^{-2} \log n)$ for $A \geq 16$; and $\mathbb{P}[\text{all cycles of } \sigma \text{ have the same length } \ell < n] \leq 4d(n)/n$, $d(n)$ the number of divisors.

Proof. (i) For $\beta \neq \ell$ the joining instances $(L, \{j, j'\})$ with j in an ℓ -cycle and j' in a β -cycle number $\ell N_\ell \beta N_\beta$ per square, the splitting instances (pairs at distance ℓ on an $(\ell + \beta)$ -cycle) number $(\ell + \beta) N_{\ell+\beta}$, and $|J| = 2|X_A| \leq 2|X|$ gives $\mathbb{E}[\ell N_\ell \beta N_\beta] \leq 2\mathbb{E}[(\ell + \beta) N_{\ell+\beta}]$; for $\beta = \ell$ the same count with $\ell^2 \binom{N_\ell}{2}$ and $\ell N_{2\ell}$ gives $\mathbb{E}[\ell^2 N_\ell (N_\ell - 1)] \leq 4\ell \mathbb{E}[N_{2\ell}]$. Summing the first over $\beta \neq \ell$ ($\sum_\beta \beta N_\beta = n$) and adding the second (its right side is the missing summand $m = 2\ell$): $\mathbb{E}[\ell N_\ell (n - \ell N_\ell)] + \mathbb{E}[\ell^2 N_\ell (N_\ell - 1)] \leq 2\mathbb{E}[\sum_{m \geq \ell+2} m N_m] \leq 2n$. Hence $n\mathbb{E}[\ell N_\ell] = \mathbb{E}[\ell N_\ell (n - \ell N_\ell)] + \mathbb{E}[\ell^2 N_\ell (N_\ell - 1)] + \ell \mathbb{E}[\ell N_\ell] \leq 2n + \ell \mathbb{E}[\ell N_\ell]$.

(ii) Count the instances $(L, (C_1, j_1), \dots, (C_k, j_k))$: C_i pairwise distinct σ -cycles of L of lengths ℓ_i and $j_i \in C_i$; there are $\ell_1 \cdots \ell_k$ of them per ordered k -tuple of distinct cycles with these lengths. Join C_1 and C_2 at the pair $\{j_1, j_2\}$ by a joining step: flip if the pair is an A -pair; if it is a B -pair, first trade the row cycle C_2 of j_2 (a switch in the sense of Lemma 2.3, whose proof applies to the row cycle through either column of the pair; it inverts C_2 in place and does not touch the cycle of j_1), then flip. The flip replaces σ by $\sigma \circ (j_1 j_2)$ on rows 1, 2, so the result has j_1, j_2 on one cycle of length $\ell_1 + \ell_2$ with $j_2 = \sigma^{\ell_2}(j_1)$ whether or not the trade was applied

(CGW Lemma 3.3), every other cycle of σ (in particular $C_3, \dots, C_k$) untouched, and the map is at most two-to-one (the backflip at $\{j_1, j_2\}$ returns L or the traded square). Repeat with $\{j_1, j_3\}, \dots, \{j_1, j_k\}$, always trading the cycle of the new column when needed, so that the cycle containing j_1 is never inverted: after step i every earlier j_r is shifted by exactly ℓ_{i+1} . After $k-1$ steps all j_i lie on one cycle of length $m = \sum \ell_i$ at the positions $j_i = \sigma^{\ell_i + \dots + \ell_k}(j_1)$, determined by j_1 and $(\ell_1, \dots, \ell_k)$, and the composite is at most 2^{k-1} -to-one. (With CGW's min-symbol rule the trade may invert the merged cycle and the positions are not determined; the count would then be 2^{2k-3} -to-one.) Hence the number of instances with lengths (ℓ_i) is at most $2^{k-1} \sum_{L'} m N_m(L')$, i.e. $\mathbb{E}[\ell_1 \dots \ell_k T(\ell_1, \dots, \ell_k)] \leq 2^{k-1} \mathbb{E}[m N_m]$ with T the number of ordered k -tuples of distinct cycles with these lengths; summing over (ℓ_i) gives the first bound, since $\sum_{(\ell_i)} T = (\kappa)_k$. For the second, split at $m \leq n/2$, where $\mathbb{E}[m N_m] \leq 4$ by (i) and $\sum_{m \leq n/2} \sum_{\sum \ell_i = m} \prod \ell_i^{-1} \leq (\sum_{\ell \leq n/2} \ell^{-1})^k \leq (\log n)^k$, and $m > n/2$, where some $\ell_i \geq m/k$, so $\sum_{\sum \ell_i = m} \prod \ell_i^{-1} \leq k \cdot (k/m)(\log n)^{k-1}$, and $\sum_{m > n/2} m N_m \cdot k^2 (\log n)^{k-1} / m \leq k^2 (\log n)^{k-1}$ because at most one cycle is longer than $n/2$.

(iii) Markov for $(\kappa)_k$ with $k = \lceil \log n \rceil \leq \log n + 1$: on $\{\kappa \geq K\}$, $(\kappa)_k \geq (K - k + 1)^k \geq ((A-1) \log n)^k$ for $K = A \log n$, and $k^2 / \log n \leq \log n + 3$, so $\mathbb{P}[\kappa \geq K] \leq 2^{k-1} (4 + k^2 / \log n) (\log n)^k / ((A-1) \log n)^k \leq (7 + \log n) (2/(A-1))^k \leq (7 + \log n) (2/(A-1))^{\log n}$ for $A \geq 3$, and $(A-1)/2 \geq e^2$ for $A \geq 16$. For the equal-length types, $\ell \mid n$ and $\ell < n$ force $\ell \leq n/2$, so $\mathbb{P}[N_\ell = n/\ell] \leq \mathbb{E}[N_\ell] \ell / n \leq 4/n$; $\ell = n$ is the n -cycle. $\square$

Theorem 3.3. Assume EQ(δ) with $\delta = \delta(n) \rightarrow 0$. Then

$$d_{\text{TV}}(P_n, Q_n) = O(\delta \log n + n^{-1+o(1)}).$$

In particular EQ($o(1/\log n)$) implies Conjecture 1.1.

Proof. Call λ plain if not all its parts are equal (so the only non-plain types are (ℓ^κ) , $\kappa \geq 2$).

Routing. Every plain λ with $\kappa = \kappa(\lambda)$ parts is reachable from (n) by a chain of $\kappa - 1$ splits, each with $\alpha \neq \beta$. Build the chain in reverse, splitting parts of λ off a shrinking residual, in a suitable order; a collision means splitting a part p off a residual R with $p = R - p$. If the two largest parts of λ are unequal, split smallest-first and let them be the final pair: while at least three parts remain, the split-off minimum p satisfies $R - p \geq 2p > p$ (all parts are ≥ 2 and at least two other parts remain), so no collision occurs, and the final split is unequal by hypothesis. If instead the two largest parts are equal, say to $\ell_{\max}$, then $\ell_{\max} \neq n/2$ (else $\lambda = (n/2, n/2)$ is not plain), so we may split one copy of $\ell_{\max}$ off first without collision, then proceed smallest-first among the remaining parts, reserving for the final split the pair $(b, \ell_{\max})$ where $b < \ell_{\max}$ is the smallest part: at every intermediate step the residual minus the split-off minimum still contains the reserved $\ell_{\max}$ and the reserved b , hence exceeds the minimum (also when a further copy of $\ell_{\max}$ is the split-off minimum: $R - p \geq \ell_{\max} + b > p$), and the final split is unequal.

For each step of such a chain, Theorem 2.4 and EQ(δ) give $|S(p_{i+1})|/|S(p_i)| = (\gamma(p_{i+1})/\gamma(p_i))(1 + \varepsilon_i)$ with $|\varepsilon_i| \leq \delta$. Multiplying along the chain,

$$\frac{P_n(\lambda)}{P_n((n))} = \frac{\gamma(\lambda)}{\gamma((n))} \prod_{i=1}^{\kappa-1} (1 + \varepsilon_i)^{\pm 1}, \quad |\varepsilon_i| \leq \delta,$$

so for all plain λ with $\kappa(\lambda) \leq K$, $P_n(\lambda) = Q_n(\lambda) Z^{-1} (1 + O(K\delta))$ (the bound is vacuous unless $K\delta = O(1)$, which is the case of interest) where $Z > 0$ is a normalising constant independent of λ (namely $Z = \sum_{\nu} \gamma(\nu) C_n(\nu) / (|\mathcal{D}_n| C_n((n)))$).

Let $K = \lceil A \log n \rceil$ with A a large absolute constant, and let $\mathcal{B}$ be the set of types that are either non-plain or have $\kappa > K$. Under Q_n , Markov's inequality with the generating function $\sum_{\pi \in \mathcal{D}_n} 2^{\kappa(\pi)} = n! [x^n] e^{-2x} (1-x)^{-2} \leq n \cdot n!$ (cf. [1, proof of Lemma 4.4]) gives $Q_n(\kappa > K) \leq e n 2^{-K} = O(n^{-2})$ for $A \log 2 \geq 3$; also $Q_n(\text{non-plain}) \leq \sum_{\ell|n, \ell < n} \gamma((\ell^n/\ell))/D_n = O(n^{-1})$. Under P_n , Proposition 3.2 gives $P_n(\kappa > K) = O(n^{-2} \log n)$ for $A \geq 16$ and $P_n(\text{non-plain}) = O(n^{-1+o(1)})$ without any input from [1] beyond the identity of Theorem 2.4.

On the complement of $\mathcal{B}$, summing $P_n = Q_n Z^{-1} (1 + O(K\delta))$ over $\lambda \notin \mathcal{B}$ and using the tail bounds shows $Z^{-1} = 1 + O(K\delta) + O(n^{-1+o(1)})$, whence $\sum_{\lambda \notin \mathcal{B}} |P_n - Q_n| = O(K\delta) + O(n^{-1+o(1)})$. Combining with the tails, $d_{\text{TV}}(P_n, Q_n) = O(\delta \log n + n^{-1+o(1)})$. $\square$

Remark 3.4 (the gap in CGW's Lemma 3.12). In [1], the direction $|S(\lambda, F)|N_\lambda \geq \frac{1}{2}|S(\mu, F)|N_\mu$ of Lemma 3.12 uses only that each splitting pair carries at most two edges and each joining pair at least one; by Theorem 2.4 it is the trivial inequality $|X| \geq |X_A|$. The direction $\leq \frac{3}{2}$ requires every splitting pair of every λ -square to carry exactly two edges that are also edges of the joining multigraph, and case 3 of the splitting procedure fails to supply them for every split other than $(2, 2)$ [2]. Consequently the upper bound of CGW's Theorem 3.13, the lower bound of their Lemma 4.1, the parity bounds of Theorem 4.3, the bound $\mathbb{P}[\sigma \text{ is an } n\text{-cycle}] \leq 2n^{-2/3}$ of Lemma 4.4, both halves of Corollary 4.5, and the results of their Section 4.2 that rest on these (Corollary 4.6, Lemma 4.7, Theorems 4.9 and 4.11, Corollary 4.10 — the last superseded by [4]) are not established by the published argument. For $\min(\alpha, \beta) = 2$ the count can be corrected with the constant 2 in place of $\frac{3}{2}$ [2, Prop. 3]: for marks at distance 2 or $m-2$ on an m -cycle, $C_n(\lambda)/C_n(\mu) \leq 2$, i.e. $\mathbb{P}_X[A] \geq \frac{1}{4}$ unconditionally. The present paper uses none of them: Proposition 3.2 replaces Corollary 4.5 in the proof of Theorem 3.3. The exact completion counts tabulated in [1] give $C_n(\lambda)/C_n(\mu) \in [0.985, 1.0035]$ for every adjacent pair at $7 \leq n \leq 11$ (and $= \frac{3}{2}$ at $n = 5$, $\frac{7}{8}$, $\frac{11}{14}$, $\frac{21}{22}$ at $n = 6$), so the statement itself is not in doubt.

4 The ladder identity

Notation: $(L, P) \in X = X(n, \lambda; \alpha, \beta)$, $P = \{j, j'\}$, frame $\pi = \pi_P$ ($L(\pi(r), j') = L(r, j)$), C_r the π -cycle of row r , $\rho_{x,y}$ the column permutation of rows x, y ($L(y, \rho_{x,y}(q)) = L(x, q)$), (x, y) *flippable* iff j, j' lie in different cycles of $\rho_{x,y}$, and the j' -trade at (x, y) exchanges rows x, y on the $\rho_{x,y}$ -cycle through j' , which replaces π by $(xy) \circ \pi$ if (x, y) is flippable and by $(xy) \circ \pi \circ (xy)$ otherwise. $B = \{1 \sim 2\}$ (rows 1, 2 in a common frame cycle), A its complement, $\mathbf{1}_B$ the bit; for B -instances d_{12} and d_{21} are the numbers of π -steps from 1 to 2 and from 2 to 1.

Definition 4.1 (ladder). Put $K_0 = \min(|C_1|, |C_2|)$ on A and $K_0 = \min(d_{12}, d_{21})$ on B , and for $1 \leq k < K_0$ let

$$x_k = \pi^{-k}(1), \quad y_k = \pi^{-k}(2), \quad T_k = \text{the } j'\text{-trade at } (x_k, y_k).$$

The pairs $\{x_k, y_k\}$ are the *ladder* of (L, P) , and $\text{Flip}(L, P) = \{k < K_0 : (x_k, y_k) \text{ flippable}\}$.

The rows $x_1, \dots, x_{K_0-1}$ are distinct rows of $C_1 \setminus \{1\}$ (on A) or of the arc from 2 to 1 (on B), the y_k likewise in $C_2 \setminus \{2\}$ or the arc from 1 to 2; so the $2(K_0 - 1)$ ladder rows are distinct and avoid rows 1, 2, and $K_0 \geq 2$: on A because π is a derangement, on B by the admissibility of the mark $(\pi(1) \neq 2, \pi(2) \neq 1)$.

Theorem 4.2 (ladder identity). *For every n , λ and split $\alpha + \beta = m$ with $\alpha, \beta \geq 2$,*

$$\mathbb{P}_X[B] - \mathbb{P}_X[A] = \mathbb{P}_X[B, \text{Flip} = \emptyset] - \mathbb{P}_X[A, \text{Flip} = \emptyset], \quad \text{hence} \quad |\mathbb{P}_X[B] - \tfrac{1}{2}| \leq \tfrac{1}{2} \mathbb{P}_X[\text{Flip} = \emptyset].$$

The same identity holds with X replaced by the set of squares with a fixed pair of rows 1, 2 (of type λ) and a fixed mark P .

Proof. Fix $(L, P) \in X$ and write $K = \{1, \dots, K_0 - 1\}$. The ladder depends on (L, P) , so the T_k are not maps of X ; the statement to prove is that $(L, P) \sim T_S(L, P)$, $S \subseteq K$ (the trades at the unordered pairs of S , in any order), is an equivalence relation on X whose classes have exactly $2^{|K|}$ elements and on which the bit is $\mathbf{1}_B(T_S(L, P)) = \mathbf{1}_B(L, P) + |S \cap \text{Flip}| \bmod 2$. This follows once we know that each T_k is an involution of X which preserves K_0 , the set of unordered ladder pairs and the flippability of each of them, that T_k toggles the bit exactly when $k \in \text{Flip}$, and that trades at disjoint row pairs commute: then the same trades, at the same unordered pairs with the same flippabilities, are available from every point of a class (symmetry and transitivity), the $2^{|K|}$ points $T_S(L, P)$ are distinct (rows x_k, y_k are changed by T_k and by no other T_i), and the bit formula holds. If $\text{Flip} \neq \emptyset$ exactly half of the subsets S have $|S \cap \text{Flip}|$ odd, so B and A each contain half of the class; summing over classes gives the identity, and the bound follows from $\mathbb{P}[B] + \mathbb{P}[A] = 1$.

T_k is a well-defined involution of X . The j' -trade at (x_k, y_k) changes only rows $x_k, y_k \notin \{1, 2\}$, so σ and the mark are untouched and (L, P) stays in X ; it is an involution because the traded columns are again the ρ'_{x_k, y_k} -cycle through j' .

Effect on the frame. If $k \in \text{Flip}$ the new frame is $\pi' = (x_k y_k) \circ \pi$. On A this merges C_1 and C_2 : the new cycle runs $1 \rightarrow \pi(1) \rightarrow \dots \rightarrow x_{k+1} \rightarrow y_k \rightarrow \pi(y_k) \rightarrow \dots \rightarrow 2 \rightarrow \dots \rightarrow y_{k+1} \rightarrow x_k \rightarrow \dots \rightarrow 1$, so $d'_{12} = |C_1|$ and $d'_{21} = |C_2|$ (reading $x_{|C_1|} = 1$, $y_{|C_2|} = 2$). On B , x_k lies on the arc from 2 to 1 and y_k on the arc from 1 to 2 (as $k < \min(d_{12}, d_{21})$), and the split into $(x_k, \pi(x_k), \dots, \pi^{-1}(y_k))$ and $(y_k, \dots, \pi^{-1}(x_k))$ puts 1 (k steps after x_k , fewer than the d_{12} steps from x_k to y_k) in the first cycle and 2 in the second: the new state is in A with $|C'_1| = d_{12}$, $|C'_2| = d_{21}$. In both cases K_0 is preserved and the bit toggles. If $k \notin \text{Flip}$ the new frame is $(x_k y_k) \circ \pi \circ (x_k y_k)$: the cycle partition is unchanged up to exchanging the labels x_k, y_k , so K_0 and the bit are preserved.

The ladder is preserved. Let $\pi' = (x_k y_k) \circ \pi$ (the conjugation case is similar and easier). Since $\pi'^{-1}(r) = \pi^{-1}((x_k y_k)(r))$, the preimage chains of 1 and 2 under π' agree with those under π up to depth k and are exchanged beyond it: $\pi'^{-i}(1) = x_i$ and $\pi'^{-i}(2) = y_i$ for $i \leq k$, while $\pi'^{-(k+1)}(1) = \pi^{-1}(y_k) = y_{k+1}$ and then $\pi'^{-i}(1) = y_i$, $\pi'^{-i}(2) = x_i$ for $k < i < K_0$. So the unordered pairs $\{x_i, y_i\}$, $i \in K$, are the ladder of $T_k(L, P)$. In the conjugation case the same computation with $(x_k y_k) \circ \pi \circ (x_k y_k)$ gives $\pi'^{-i}(1) = (x_k y_k)(x_i)$, which again permutes the pairs $\{x_i, y_i\}$ among themselves (only $i = k$ is affected, by exchanging x_k, y_k).

Flippability is preserved and the T_k commute. T_k changes rows x_k, y_k only, and the flippability of (x_i, y_i) , $i \neq k$, depends only on rows x_i, y_i , which are different rows. Likewise T_i and T_k act on disjoint pairs of rows, each along a column set determined by its own two rows, so $T_i T_k = T_k T_i$. (After T_k the pair $\{x_i, y_i\}$ may have changed its index, but the trade at an unordered pair does not depend on the index or on the order of the two rows.)

The statement for fixed R holds because every T_k fixes rows 1 and 2. □

Corollary 4.3 (the conjecture reduces to the ladder). *Write $f(n, \lambda; \alpha, \beta) = \mathbb{P}_X[\text{Flip} = \emptyset]$. By Theorem 2.4, $\bar{q}(\mu \rightarrow \lambda) - 1 = (\mathbb{P}_X[B] - \mathbb{P}_X[A]) / \mathbb{P}_X[A]$ and $\mathbb{P}_X[A] \geq (1 - f)/2$, so $|\bar{q} - 1| \leq$*

$2f/(1-f)$: EQ(δ) holds with $\delta(n) = \max_{\lambda, \alpha \neq \beta} 2f/(1-f)$. If this $\delta(n) \rightarrow 0$, Theorem 3.3 gives

$$d_{\text{TV}}(P_n, Q_n) = O\left(\log n \cdot \max_{\lambda, \alpha \neq \beta} f(n, \lambda; \alpha, \beta) + n^{-1+o(1)}\right).$$

In particular the weak CGW conjecture follows from

$$(L) \quad \mathbb{P}_X[\text{no ladder pair is flippable}] = o(1/\log n) \quad \text{uniformly in } \lambda, \alpha, \beta. \quad (2)$$

Remark 4.4 (data). On Jacobson–Matthews samples with a uniformly random mark (9000, 7200, 3000 ladder-pair instances at $n = 30, 50, 100$; runs/s13/f1/): $\mathbb{P}[\text{Flip} = \emptyset \mid K_0] = 2^{-(K_0-1)}$ within statistical error for $K_0 \leq 6$ and consistent with it beyond; $\mathbb{P}[\text{Flip} = \emptyset] = 0.125, 0.079, 0.041 \approx 4/n$; the ladder pairs are flippable with frequency 0.5002 (62082 pairs at $n = 50$); and $\mathbb{P}[B] - \mathbb{P}[A] = -0.022 \pm 0.011, +0.008 \pm 0.012, -0.030 \pm 0.018$.

4.1 Offset ladders and short arcs

The ladder is one of a family. For an integer c (an *offset*) put

$$K_c = \begin{cases} \min(|C_1| - c, |C_2|) & \text{on } A, \\ \min(d_{21} - c, d_{12}) & \text{on } B, \end{cases} \quad D_c = \{(x_{c+i}, y_i) = (\pi^{-(c+i)}(1), \pi^{-i}(2)) : 1 \leq i < K_c\},$$

so D_0 is the ladder and for $c > 0$ the x -row is c steps deeper than the y -row (x -depths $c+1, \dots, c+K_c-1$ stay inside $C_1 \setminus \{1\}$, resp. the arc from 2 to 1; y -depths $1, \dots, K_c-1$ inside $C_2 \setminus \{2\}$, resp. the arc from 1 to 2). We take $c \geq 0$; the families with the y -row deeper are obtained by exchanging the roles of rows 1 and 2 (and of C_1, C_2, d_{12}, d_{21}) throughout.

Theorem 4.5 (offset ladders). *For every offset $c \geq 0$, on the domain $\mathcal{D}_c = \{K_c \geq 2\}$ (where $D_c \neq \emptyset$) the j' -trades at the pairs of D_c are commuting involutions that preserve K_c , the unordered pairs of D_c and their flippabilities, and toggle the bit iff the traded pair is flippable. Hence, with Flip_c the set of flippable pairs of D_c ,*

$$\mathbb{P}[B, \mathcal{D}_c] - \mathbb{P}[A, \mathcal{D}_c] = \mathbb{P}[B, \mathcal{D}_c, \text{Flip}_c = \emptyset] - \mathbb{P}[A, \mathcal{D}_c, \text{Flip}_c = \emptyset].$$

Proof. Identical to Theorem 4.2; only the bookkeeping changes. Let $\pi' = (x_{c+i} y_i) \circ \pi$. The preimage chain of 1 under π' agrees with that of π up to depth $c+i$ and then continues along the chain of 2: $\pi'^{-(c+i+t)}(1) = y_{i+t}$; symmetrically $\pi'^{-(i+t)}(2) = x_{c+i+t}$ ($t \geq 1$). So the pair of D_c at y -depth $i' > i$ is $\{y_{i'}, x_{c+i'}\}$, unchanged as a set, and the pairs at depth $i' < i$ are untouched: D_c is preserved. (The same computation shows that trading a pair at depths (k, k') sends a deeper pair at depths (K, K') to depths $(K' + k - k', K - k + k')$, whose offset is $2(k - k') - (K - K')$; it is preserved iff $K - K' = k - k'$, so only families of constant offset, i.e. the D_c , are preserved.) On A the trade merges C_1, C_2 into a cycle with $d'_{12} = |C_1| - (c+i) + i = |C_1| - c$ and $d'_{21} = |C_2| + c$, so $K'_c = \min(d'_{21} - c, d'_{12}) = \min(|C_2|, |C_1| - c) = K_c$; on B the pair separates rows 1, 2 ($c+i < d_{21}$ puts x_{c+i} on the arc from 2 to 1, $i < d_{12}$ puts y_i on the arc from 1 to 2) and gives $|C'_1| = d_{12} + c$, $|C'_2| = d_{21} - c$, again $K'_c = \min(d_{12}, d_{21} - c) = K_c$. No pair of D_c is frame-adjacent: $y_i = \pi(x_{c+i})$ would need $d_{21} = c - 1$ (or $\pi(1) = 2$ when $c = 0$) and $x_{c+i} = \pi(y_i)$ would need $d_{21} = c + 1$, both excluded on $\mathcal{D}_c$. Commutation, preservation of flippability and the orbit count are as before. $\square$

Exact enumeration at $n = 7$ ($\lambda = (7)$, $\{\alpha, \beta\} = \{2, 5\}$) confirms the identities for $c = 0, 1, 2, 3$ to all digits ($\mathbb{P}[B, \mathcal{D}_c] - \mathbb{P}[A, \mathcal{D}_c] = -0.005848, +0.001462, +0.003509, -0.001316$, equal to the right-hand sides).

Using the offsets to price short arcs. Fix $\ell \geq 2$ and consider B -instances with $d_{12} = \ell$. For $0 \leq c \leq n/2 - \ell$ and $d_{21} \geq c + \ell$ one has $K_c = \ell$, so D_c has $\ell - 1$ pairs, and the offset- c trade at a flippable pair of D_c lands in A with $|C_1| = \ell + c \leq n/2$ and $|C_2| = d_{21} - c \geq \ell$. Within a class of $\mathcal{D}_c$ all B -members have the same (d_{12}, d_{21}) and all A -members the same $(|C_1|, |C_2|) = (d_{12} + c, d_{21} - c)$, and Flip_c is a class invariant, so Theorem 4.5 class by class and Proposition 4.6 give

$$\mathbb{P}[B, d_{12} = \ell, d_{21} \geq c + \ell, \text{Flip}_c \neq \emptyset] = \mathbb{P}[A, |C_1| = \ell + c, |C_2| \geq \ell, \text{Flip}_c \neq \emptyset] \leq \frac{18}{n - \ell - c - 1} \leq \frac{36}{n - 2}. \quad (3)$$

Averaging over c : if $\theta(L, P)$ denotes the fraction of offsets $c \in [0, n/2 - \ell]$ whose diagonal D_c contains a flippable pair, then for every $\theta_0 > 0$

$$\mathbb{P}[B, d_{12} = \ell, |C_1| \geq n/2 + \ell, \theta \geq \theta_0] \leq \frac{36}{(n - 2)\theta_0}, \quad (4)$$

since on $|C_1| \geq n/2 + \ell$ one has $d_{21} = |C_1| - \ell \geq n/2 \geq c + \ell$ for every such c , so $\mathbb{E}[\theta \mathbf{1}_E]$ is the average over c of the left side of (3), and Markov's inequality applies. Together with the splice-in bound $\mathbb{P}[B, d_{12} = \ell, |C_1| < n/2 + \ell] \leq 4/(n - 2\ell + 1) \leq 8/n$ for $\ell \leq n/4$ (Proposition 4.7 below), this prices every short arc except on the event $\{\theta < \theta_0\}$.

Proposition 4.6 (short frame cycles through a marked row are rare). *For (L, P) uniform on X and every $2 \leq \ell \leq n/2$,*

$$\mathbb{P}[|C_1| = \ell] \leq \frac{18}{n - \ell - 1}, \quad \text{hence} \quad \mathbb{P}[|C_1| \leq \ell] \leq \frac{36\ell}{n} \quad (\ell \leq n/2 - 1),$$

and the same bounds hold for C_2 .

Proof. Let $E_\ell = \{(L, P) \in X : |C_1| = \ell\}$. (The case $C_1 = \{1, 2\}$ cannot occur: it would mean $\sigma(j) = j'$ and $\sigma(j') = j$, but $j' = \sigma^\alpha(j)$ lies in an m -cycle with $m \geq 4$.)

Forward moves. From $(L, P) \in E_\ell$ choose $y \in C_1 \setminus \{1, 2\}$ and $x \notin C_1 \cup \{2\}$: at least $(\ell - 2)(n - \ell - 1)$ pairs $((\ell - 1)(n - \ell - 1)$ if $2 \notin C_1$; for $\ell = 2$ this is $n - 3$). If (x, y) is flippable, apply the j' -trade at (x, y) (trade rows x, y along the $\rho_{x,y}$ -cycle $Q_{x,y}$ through j'); otherwise turn C_1 (which makes (x, y) flippable, as $y \in C_1 \not\equiv x$) and then trade $Q_{x,y}$ in the turned square. In both cases the resulting frame is $(xy) \circ \pi^\pm$ with $\pi^\pm$ equal to π or to π inverted on C_1 , so the cycle of row 1 becomes $C_1 \cup C_x$, of length $\ell + |C_x| > \ell$; rows 1, 2 are untouched by the trade, and the result is in X (no turn, or $2 \in C_1$) or in J (turn with $2 \notin C_1$).

Backward moves. A square $(L', P) \in X \cup J$ is reached from $(L, P) \in E_\ell$ by the move at (x, y) only if x, y lie in the cycle C'_1 of row 1 in L' , and undoing the trade (the same trade, since $\rho'_{x,y} = \rho_{x,y}^{-1}$ on $Q_{x,y}$) splits C'_1 into the cycles $\{x, \pi'(x), \dots, \pi'^{-1}(y)\}$ and $\{y, \pi'(y), \dots, \pi'^{-1}(x)\}$ of $(xy) \circ \pi'$, of which the one containing row 1 must have length exactly ℓ : given L' , that arc is determined by its position relative to row 1 (ℓ choices), and the arc determines $\{x, y\}$. With the binary choice “turn back or not” this gives at most 2ℓ preimages per (L', P) .

Counting. The map $((L, P), \text{move}) \mapsto ((L', P), \text{inverse move})$ is injective, so $|E_\ell|(\ell - 2)(n - \ell - 1) \leq 2\ell(|X| + |J|) \leq 6\ell|X|$, using $|J| = 2|X_A| \leq 2|X|$ (Theorem 2.4). For $\ell \geq 3$ this is the claim; for $\ell = 2$ the forward count is $n - 3$ and the backward count 4, giving $12/(n - 3)$. The bound for C_2 is proved by the same argument with the roles of rows 1 and 2 exchanged: a turn of $C_2 \not\equiv 1$ replaces σ by $(\sigma(j)\sigma(j')) \circ \sigma$, which also splits the m -cycle into lengths α, β and lands in J . $\square$

Proposition 4.7 (the splice-in bound). *For $2 \leq \ell \leq n/4$, $\mathbb{P}_X[B, d_{12} = \ell, |C_1| < n/2 + \ell] \leq 4/(n - 2\ell + 1) \leq 8/n$, and the same with d_{21} in place of d_{12} .*

Proof. Let E be the event. *Forward moves.* From $(L, P) \in E$ choose $y = \pi^i(1)$ with $1 \leq i \leq \ell - 1$ (a row strictly inside the arc from 1 to 2; $\ell - 1$ choices) and a row $x \notin C_1$ ($n - |C_1| \geq (n - 2\ell + 1)/2$ choices), and let C_x be the frame cycle of x , which contains neither 1 nor 2. If (x, y) is not flippable, turn C_x : this trades columns j, j' on the rows of C_x only, so it preserves X , inverts π on C_x (Lemma 2.2), fixes row y , and replaces $\rho_{x,y}$ by $\rho_{x,y} \circ (j j')$, which makes (x, y) flippable. Then apply the j' -trade at (x, y) , which replaces the frame by $(x y) \circ \pi^\pm$ ($\pi^\pm = \pi$, or π inverted on C_x): the arc from 1 to 2 becomes $1 \rightarrow \dots \rightarrow \pi^{-1}(y) \rightarrow x \rightarrow \dots \rightarrow y \rightarrow \dots \rightarrow 2$ with C_x occupying positions $i, \dots, i + |C_x| - 1$, so d_{12} grows by $|C_x|$, rows 1, 2 are untouched, and the result is again in X_B with the same mark. *Backward moves.* Given the target $(L', P) \in X_B$ and ℓ , a preimage is determined by the position $i \in \{1, \dots, \ell - 1\}$ of x on the arc of L' from 1 to 2 (then $|C_x| = d'_{12} - \ell$ and $y = \pi'^{i+|C_x|}(1)$ are determined), together with the binary choice of whether C_x was turned; undoing is the same j' -trade (the pair stays flippable, $\rho'_{x,y} = \rho_{x,y}^{-1}$ on the traded cycle) followed by the turn. So each target has at most $2(\ell - 1)$ preimages while each source has at least $(\ell - 1)(n - 2\ell + 1)/2$ forward moves, whence $|E|(\ell - 1)(n - 2\ell + 1)/2 \leq 2(\ell - 1)|X|$. $\square$

5 The orbit method

Throughout this section $X = X(n, \lambda; \alpha, \beta)$, and “legal” means: preserves the type of rows 1, 2 and the mark, i.e. maps X to itself. The mark is now written $\{p, p'\}$ (so $p = j$, $p' = j'$ in the notation of §§2–4; the letters j, j' are freed for other columns), π is its frame, and a column pair (q, c) is *free* if $\{q, c\} \cap \{p, p'\} = \emptyset$. A pair of rows (x, y) is called *parallel* if it is flippable (p, p' in different cycles of $\rho_{x,y}$) and *crossed* otherwise; Q_p and $Q_{p'}$ denote the $\rho_{x,y}$ -cycles through p and p' . A free pair (q, c) is *separated* by $\{p, p'\}$ in $\rho_{x,y}$ if q and c lie on different arcs of a common cycle through p, p' (crossed case) or one in Q_p and the other in $Q_{p'} \neq Q_p$ (parallel case).

(a) Transposition: which half of the conditioning is the obstacle. Let L be uniform on *all* Latin squares of order n (no condition on rows 1, 2 in this paragraph), let x, y carry four distinct symbols on $\{p, p'\}$, and condition on the two full columns p, p' (the frame π of (p, p')). In the transposed square $L^\top$ the columns p, p' are the distinguished rows, the frame π is their column permutation, and the event “ p, p' lie in different cycles of $\rho_{x,y}$ ” is the event “ $\{x, y\}$ is an A -pair for the distinguished rows”. Conditioning on the two full rows p, p' of $L^\top$ is conditioning on their type and on the position of the pair $\{x, y\}$ in their cycle structure, which is free by symbol and column relabelling. Hence, by Lemma 2.3 ($|J_A| = |J_B|$, which needs no condition on the cycle lengths) and CGW’s Theorem 3.13 (the inequality $\mathbb{P}_X[A] \geq \frac{1}{3}$, whose published proof has a gap for every split other than $(2, 2)$, Remark 3.4):

$$\mathbb{P}[p, p' \text{ in different cycles of } \rho_{x,y} \mid \text{columns } p, p'] \begin{cases} = \frac{1}{2} & \text{if } x, y \text{ are apart in the frame,} \\ \geq \frac{1}{3} & \text{if together at distances } \geq 2. \end{cases} \quad (5)$$

(The second bound is CGW’s $\frac{3}{2}$, affected by the gap; it holds with $\frac{1}{4}$ unconditionally at distance 2, Remark 3.4.) So the statement with the *columns* prescribed is CGW’s theorem; it is the conditioning on rows 1, 2 — on their type and the mark, which is what defines X and cannot be removed — that is the obstacle. In the same transposed picture the ladder trades T_k of Theorem 4.2 at flippable pairs are CGW’s flips and backflips at the column pairs $\{x_k, y_k\}$ of

$L^\top$, and Theorem 4.2 is exactly the legal half of CGW's Lemma 3.12 for the distinguished rows p, p' : the other half — the switch and the cross-switch at $\{x_k, y_k\}$ — trades the columns p, p' of L along a frame cycle or frame arc through exactly one of rows 1, 2 (apart: $C_1 \ni 1$, $C_1 \not\ni 2$; together: the arc from x_k to y_k contains 1 but not 2, the other arc 2 but not 1), so it changes the type of rows 1, 2 and leaves X . The switch is, in the original orientation, the unflip of rows 1, 2 at (p, p') (which maps $X_A \rightarrow J_A$); it toggles the crossed/parallel status of *every* pair (x, y) with x on the side of row 1 and y on the side of row 2, and reverses the orientation of one side, so it carries the ladder (a diagonal of the side-by-side array of pairs) to a different family (an anti-diagonal). The cross-switch likewise maps X_B to J_B (it splits the σ -cycle through p, p' while keeping rows 1, 2 in one frame cycle, CGW Lemma 3.10) and toggles the interior pairs. Both are exact but not usable for the ladder bits in X .

(b) Row trades are free in X ; column turns are conditionally free. For rows $z, w \notin \{1, 2\}$ every trade of rows z, w along a $\rho_{z, w}$ -cycle or $\rho_{z, w}$ -arc changes rows z, w only, hence is legal in X . In particular:

Lemma 5.1 (exact fairness of column-pair status for generic rows). *Let $x, y \notin \{1, 2\}$ and let (q, c) be a column pair lying in two different cycles of $\rho_{x, y}$. The CGW switch at (q, c) for the rows x, y (trade of rows x, y along the one of the two cycles that contains the smaller minimum symbol) is a legal involution of X that toggles whether x, y lie in one column cycle of (q, c) , and preserves the cycle partition of $\rho_{x, y}$ and the separation status of every column pair with respect to $\{p, p'\}$. Hence, conditionally on everything determined by the other rows and by the cycle partition of $\rho_{x, y}$,*

$$\mathbb{P}_X[x, y \text{ in different column cycles of } (q, c)] = \frac{1}{2}.$$

Proof. As in Lemma 2.3 with $(1, 2)$ replaced by (x, y) : the trade swaps the entries of rows x, y on the columns of the chosen cycle $Q \ni q$, $Q \not\ni c$, so $\delta_{q, c} \mapsto \delta_{q, c} \circ (xy)$, which splits or joins the column cycles through x, y . It inverts Q in place, so the symbol sets of the two cycles are unchanged and the min-symbol rule re-selects Q : an involution. Rows 1, 2 are untouched, so type and mark are preserved. Inverting Q exchanges the two arcs between p and p' when $Q \ni p, p'$, so separation of any pair by $\{p, p'\}$ is preserved. (If Q contains p or p' the frame changes at rows x, y ; so frame-determined data are part of the conditioning only when Q avoids p, p' .) $\square$

A column turn, by contrast, changes the rows of a column cycle. Let (q, c) be a free column pair and Z one of its column cycles.

Lemma 5.2 (legal column turns). *The turn of Z (swapping the entries of columns q, c in the rows of Z) is a measure-preserving involution of X if and only if $|Z \cap \{1, 2\}| \in \{0, 2\}$; when $|Z \cap \{1, 2\}| = 2$ it conjugates σ by (qc) and keeps the type and the mark. It preserves the columns p, p' (hence the frame and every frame-determined family of row pairs) and the column-cycle partition of (q, c) and of every column pair disjoint from $\{q, c\}$ (in particular of every pair of a matching containing (q, c)). For a row pair (x, y) with $|Z \cap \{x, y\}| = 1$ it replaces $\rho_{x, y}$ by $\rho_{x, y} \circ (qc)$ (if $x \in Z$) or $(qc) \circ \rho_{x, y}$ (if $y \in Z$), which toggles the crossed/parallel status of (x, y) exactly when (q, c) is separated by $\{p, p'\}$ in $\rho_{x, y}$ (different arcs when crossed, different cycles $Q_p, Q_{p'}$ when parallel), and leaves (q, c) separated afterwards; with $|Z \cap \{x, y\}| \in \{0, 2\}$ the status is unchanged. Turns of cycles of column pairs from a fixed matching $\mathcal{M}$ of the free columns pairwise commute.*

Proof. A column cycle is a Latin trade (Lemma 2.2), so the turn is a bijection of Latin squares and an involution, and it preserves the cycle partition of (q, c) ; column pairs disjoint from $\{q, c\}$ are untouched, and so are columns p, p' . Rows 1, 2 change iff Z meets them; if $Z \ni 1, 2$ both rows swap their entries at q, c , so $\sigma' = (qc)\sigma(qc)$, whose type is that of σ and in which p, p' (fixed by (qc)) keep their distance. If Z meets exactly one of rows 1, 2, $\sigma' = \sigma \circ (qc)$ or $(qc) \circ \sigma$, which changes the type. The effect on $\rho_{x,y}$: if $x \in Z$ and $y \notin Z$ the entries of row x at q, c are exchanged, so $\rho'_{x,y} = \rho_{x,y} \circ (qc)$, and symmetrically; multiplying by a transposition of two points on one cycle splits it at those points, so p, p' are separated iff q, c lie on different arcs between them, and multiplying by a transposition of two points on different cycles merges the cycles, so p, p' are joined iff q, c lie one in each of their cycles; in both cases q, c lie one on each side afterwards. Turns of cycles of disjoint column pairs act on disjoint sets of cells; two cycles of the same pair have disjoint row sets. $\square$

(c) The product bound. Fix a matching $\mathcal{M}$ of the free columns, independent of the square, and a frame-determined family $\mathcal{G}$ of row pairs (x, y) , $x, y \notin \{1, 2\}$, whose x -rows are pairwise distinct and disjoint from its y -rows (a block of the ladder, where the pairs are even disjoint; or a union of diagonals D_c with distinct x -rows, which may share y -rows). For $(x, y) \in \mathcal{G}$ call a column pair $(q, c) \in \mathcal{M}$ a *clean coordinate* of (x, y) if the column cycle Z of (q, c) through x has length at most ℓ_1 (a fixed constant), satisfies $|Z \cap \{1, 2\}| \in \{0, 2\}$, and contains no row of $\mathcal{G}$ other than x (no other x -row and no y -row, in particular not y). Let $\mathcal{T}_{x,y}$ be the set of clean coordinates of (x, y) and $\eta_{x,y} \in \{0, 1\}^{\mathcal{T}_{x,y}}$. By Lemma 5.2 the turns of all clean coordinates of all pairs of $\mathcal{G}$ commute, generate a group acting freely on X , preserve $\mathcal{G}$ and the sets $\mathcal{T}_{x,y}$, and

$$\rho_{x,y}(\eta) = \rho_{x,y}(0) \circ \prod_{(q,c) \in \mathcal{T}_{x,y}: \eta_{x,y}(q,c)=1} (qc),$$

the bit $b_{x,y}(\eta) = [(x, y) \text{ parallel}]$ depending on $\eta_{x,y}$ only. Let the *orbit weight* of (x, y) be

$$\bar{y}_{x,y} = \mathbb{E}_{\eta_{x,y}} \left[\frac{\#\{(q, c) \in \mathcal{T}_{x,y} : (q, c) \text{ separated by } \{p, p'\} \text{ in } \rho_{x,y}(\eta)\}}{|\mathcal{T}_{x,y}|} \right]$$

(0 if $\mathcal{T}_{x,y} = \emptyset$), a function of the orbit, and $Y_{\mathcal{G}} = \sum_{(x,y) \in \mathcal{G}} \bar{y}_{x,y}$.

Proposition 5.3 (product bound). *Conditionally on the orbit, the bits $b_{x,y}$, $(x, y) \in \mathcal{G}$, are independent, and*

$$\mathbb{P}[(x, y) \text{ crossed} \mid \text{orbit}] \leq 1 - \frac{1}{2} \bar{y}_{x,y}, \quad \mathbb{P}[\text{no pair of } \mathcal{G} \text{ is flippable} \mid \text{orbit}] \leq \prod_{(x,y) \in \mathcal{G}} (1 - \frac{1}{2} \bar{y}_{x,y}) \leq e^{-Y_{\mathcal{G}}/2}.$$

Consequently $\mathbb{P}_X[\text{Flip}_{\mathcal{G}} = \emptyset] \leq \mathbb{E}_X[e^{-Y_{\mathcal{G}}/2}]$.

Proof. The action is free: distinct column pairs act on disjoint cells, and two clean coordinates of different pairs at the same (q, c) are the cycles through two distinct x -rows, each avoiding the other, hence different cycles; the sets $\mathcal{T}_{x,y}$ are orbit-invariant by Lemma 5.2 (row sets of the cycles, their meeting $\{1, 2\}$, and the frame-determined family are preserved). The uniform measure on X restricted to an orbit is therefore uniform on the orbit, which is the product of the cubes $\{0, 1\}^{\mathcal{T}_{x,y}}$. A clean coordinate of (x, y) contains x and not y , and a clean coordinate of another pair contains neither x nor y (it avoids all x -rows but its own and all y -rows), so $b_{x,y}$ depends on $\eta_{x,y}$ alone, whence the independence. Fix (x, y) and $Z = (q, c) \in \mathcal{T}_{x,y}$. Whether Z is

separated in $\rho_{x,y}(\eta)$ does not depend on η_Z (Lemma 5.2: the turn of Z leaves Z separated), and when it is, the two values of η_Z give opposite bits. Hence $\mathbb{P}[b_{x,y} = 0] \leq 1 - \frac{1}{2}\mathbb{P}_\eta[Z \text{ separated}]$ for every Z , and averaging the right side over $Z \in \mathcal{T}_{x,y}$ gives the first bound; the second is the product of the first over the independent bits. $\square$

(c') Adaptive coordinates: the toggling is automatic. Proposition 5.3 needs the clean coordinates of a ladder pair to be *separated* by $\{p, p'\}$ with positive probability, a genericity input we cannot prove (§6). It disappears if the coordinates are chosen along $\rho_{x,y}$ from the mark. For a pair (x, y) of the family and $t \geq 1$ let

$$(q^t, c^t) := (\rho_{x,y}^t(p), \rho_{x,y}^t(p')),$$

and let $\mu_{x,y} := \min(d(p \rightarrow p'), d(p' \rightarrow p))$ if (x, y) is crossed and $\min(|Q_p|, |Q_{p'}|)$ if parallel. For $t < \mu_{x,y}$ the pair (q^t, c^t) is free ($q^t \neq c^t$, both $\notin \{p, p'\}$) and separated by $\{p, p'\}$ in $\rho_{x,y}$ by construction: on different arcs when crossed, one in Q_p and one in $Q_{p'}$ when parallel. So the turn of its column cycle Z through x (with $y \notin Z$) toggles the status of (x, y) whenever it is legal (Lemma 5.2). What makes these coordinates usable in the orbit method is that they are invariant under their own turn. (We keep (c) for comparison: its orbit weight is what the data of (d) measure, and it is where the genericity constant of §6 lives.)

Lemma 5.4 (the adaptive coordinate is orbit-invariant). *Let $t < \mu_{x,y}$, let Z be the column cycle of (q^t, c^t) through x with $y \notin Z$, and let L' be the turn of Z . Then $\rho'_{x,y} = \rho_{x,y} \circ (q^t c^t)$, the status of (x, y) toggles, $\mu'_{x,y} = \mu_{x,y}$, and $(\rho'^s(p), \rho'^s(p')) = (q^s, c^s)$ for every $s \leq t$; in particular the t -th candidate of L' is (q^t, c^t) again, and its column cycle through x is Z (same row set).*

Proof. Write $\rho = \rho_{x,y}$, $q = q^t$, $c = c^t$. The turn exchanges the entries of x at q, c (and those of the other rows of Z , which do not affect ρ), so $\rho' = \rho \circ (qc)$: $\rho'(q) = \rho(c)$, $\rho'(c) = \rho(q)$, $\rho' = \rho$ elsewhere. Crossed case, with $p = u_0 \rightarrow u_1 \rightarrow \dots \rightarrow u_a = p' \rightarrow v_1 \rightarrow \dots \rightarrow v_b = p$ and $q = u_t$, $c = v_t$ ($t < \min(a, b)$): the cycles of ρ' through p and p' are $u_0 \rightarrow \dots \rightarrow u_t \rightarrow v_{t+1} \rightarrow \dots \rightarrow v_b = p$ and $u_{t+1} \rightarrow \dots \rightarrow u_a = p' \rightarrow v_1 \rightarrow \dots \rightarrow v_t \rightarrow u_{t+1}$, of lengths b and a , so (x, y) is now parallel with $\mu' = \min(a, b) = \mu$, and the paths $p \rightarrow u_s$, $p' \rightarrow v_s$ for $s \leq t$ are untouched. Parallel case, $Q_p = (p = u_0, u_1, \dots, u_{a-1})$, $Q_{p'} = (p' = v_0, \dots, v_{b-1})$, $q = u_t$, $c = v_t$ ($t < \min(a, b)$): the single cycle of ρ' is $u_0 \rightarrow \dots \rightarrow u_t \rightarrow v_{t+1} \rightarrow \dots \rightarrow v_{b-1} \rightarrow v_0 = p' \rightarrow v_1 \rightarrow \dots \rightarrow v_t \rightarrow u_{t+1} \rightarrow \dots \rightarrow u_{a-1} \rightarrow u_0$, with $d'(p \rightarrow p') = b$, $d'(p' \rightarrow p) = a$, so crossed with $\mu' = \mu$, and again the paths up to step t are untouched. The row set of the turned cycle is preserved (Lemma 2.2). $\square$

Proposition 5.5 (orbit bound with adaptive coordinates). *Let $(L, P) \in X$, I a frame-determined family of pairs (x_k, y_k) as in §(c) (distinct x -rows, disjoint from the y -rows, none in $\{1, 2\}$; the ladder, or a union of offset diagonals), and $T, \ell_1 \geq 1$. Call the candidate (k, t) good if $t \leq \min(T, \mu_k - 1)$ and the column cycle Z_k^t of (q_k^t, c_k^t) through x_k has length $\leq \ell_1$, meets $\{1, 2\}$ in 0 or 2 rows and meets the rows of I only in x_k . Let $t_0(k)$ be the least good t (if any), $C_k = \{q_k^t, c_k^t : t \leq t_0(k)\}$ (all candidates $t \leq \min(T, \mu_k - 1)$ if there is none), and*

$$U = \{k \in I : t_0(k) \text{ exists and } \{q_k^{t_0(k)}, c_k^{t_0(k)}\} \cap C_l = \emptyset \text{ for all } l \in I \setminus \{k\}\}.$$

Then the turns T_k of $Z_k^{t_0(k)}$, $k \in U$, are legal, commute, generate a free action of $\{0, 1\}^U$ on X that preserves the frame, the family, every μ_l , $t_0(l)$, C_l ($l \in I$) and U ; T_k toggles the status of (x_k, y_k) and fixes that of every other pair. Hence

$$\mathbb{P}_X[\text{Flip}_I = \emptyset] \leq \mathbb{E}_X[2^{-|U|}].$$

Proof. T_k changes only columns $q_k, c_k := q_k^{t_0(k)}, c_k^{t_0(k)}$ in the rows of Z_k (lengths of cycles are preserved with their row sets, so the length condition is invariant along with the rest). Legality: Z_k meets $\{1, 2\}$ in 0 or 2 rows and $\{q_k, c_k\} \cap \{p, p'\} = \emptyset$, so the type of rows 1, 2 and the mark are preserved (Lemma 5.2); columns p, p' are untouched, so the frame and the ladder are preserved. For $k \neq l$ in U the column pairs $\{q_k, c_k\}, \{q_l, c_l\} \subseteq C_l$ are disjoint, so the turns act on disjoint cells and commute, and the action is free. Invariance of the data of a pair $l \neq k$ under T_k : rows $x_l, y_l \notin Z_k$, so ρ_l , its status, μ_l and all its candidates are unchanged; the goodness of candidate (l, t) depends on the column cycle of (q_l^t, c_l^t) through x_l , i.e. on the contents of columns q_l^t, c_l^t , which are not in $\{q_k, c_k\}$ for $t \leq t_0(l)$ (resp. for all candidates if $t_0(l)$ does not exist) by the defining condition of U ; hence $t_0(l)$ and C_l are unchanged. Invariance of the data of k itself: by Lemma 5.4, μ_k and the candidates $t \leq t_0(k)$ are unchanged; the candidates $t < t_0(k)$ have columns different from q_k, c_k (distinct points of the two paths), so their cycles and non-goodness are unchanged, and Z_k itself keeps its row set, so $t_0(k)$ is still good; C_k is unchanged. If $Z_k \ni 1, 2$, rows 1, 2 change at q_k, c_k , which affects none of the above. Hence U is invariant, every orbit of the cube is a cube of size $2^{|U|}$ on which the status of pair $k \in U$ is $\text{bit}_k(L) \oplus \varepsilon_k$, and $\text{Flip}_I = \emptyset$ forces one value of $(\varepsilon_k)_{k \in U}$; X is uniform. $\square$

Hypothesis 5.6 (short clean cycles at the ladder rows). *There are $T, \ell_1 = O(\log^2 n)$ such that, uniformly in λ, α, β , with $\ell_0 = n/\log^2 n$ and U as in Proposition 5.5:*

- (i) *for the sub-ladder I of the first $\min(K_0 - 1, \lceil n/\log^2 n \rceil)$ ladder pairs, $\mathbb{E}_X[2^{-|U|}; K_0 \geq \ell_0] = o(1/\log n)$;*
- (ii) *for a B -instance with $d_{12} = \ell < \ell_0$ and $|C_1| \geq n/2 + \ell$, with $S = \{2\ell t : 0 \leq t \leq (n/2 - \ell)/(2\ell)\}$ (offsets at gaps 2ℓ , so that the x -rows $\pi^{-(c+i)}(1)$, $c \in S$, $1 \leq i < \ell$, are pairwise distinct and $|S| \geq n/(8\ell)$) and U taken for the family $\bigcup_{c \in S} D_c$: the event that fewer than half of the diagonals D_c , $c \in S$, meet U has probability $o(1/\log n)$ (as a single event over all $\ell < \ell_0$); and symmetrically for d_{21} .*

Proposition 5.7. *Hypothesis 5.6 implies (L), hence the weak CGW conjecture.*

Proof. By Theorem 4.2, $|\mathbb{P}[B] - \mathbb{P}[A]| \leq \mathbb{P}[\text{Flip} = \emptyset] \leq \mathbb{P}[K_0 \geq \ell_0, \text{Flip} = \emptyset] + \mathbb{P}[A, K_0 < \ell_0] + \mathbb{P}[B, K_0 < \ell_0]$. *Long ladders.* K_0 is a function of the frame, hence orbit-invariant, and by Proposition 5.5 $\mathbb{P}[\text{Flip}_I = \emptyset \mid \text{orbit}] \leq 2^{-|U|}$, so $\mathbb{P}[K_0 \geq \ell_0, \text{Flip} = \emptyset] \leq \mathbb{E}[2^{-|U|}; K_0 \geq \ell_0] = o(1/\log n)$ by (i). *Short ladders on A .* $\mathbb{P}[A, K_0 < \ell_0] \leq 72\ell_0/n$ by Proposition 4.6. *Short arcs on B .* Fix $\ell < \ell_0$ and consider B -instances with $d_{12} = \ell$. The part $|C_1| < n/2 + \ell$ costs at most $8/n$ by Proposition 4.7. On $|C_1| \geq n/2 + \ell$ apply Proposition 5.5 to the family $\bigcup_{c \in S} D_c$: its x -rows lie in the disjoint depth intervals $[c + 1, c + \ell - 1]$, $c \in S$, on the frame arc from 2 to 1, and its y -rows $\pi^{-i}(2)$ on the arc from 1 to 2, so the family is as required. Given the orbit the statuses of all pairs are independent and T_k toggles pair $k \in U$, so a diagonal D_c with $U \cap D_c \neq \emptyset$ is flippable (contains a parallel pair) with conditional probability $\geq \frac{1}{2}$, and the diagonals' flippabilities are independent (they are disjoint sets of pairs). Off the exceptional event of (ii), at least $|S|/2$ diagonals meet U , so the number of flippable offsets $c \in S$ has conditional mean $\mu \geq |S|/4$, and by Chernoff it is $\geq |S|/8$ except with conditional probability $\leq e^{-\mu/8} \leq e^{-|S|/32} \leq e^{-\log^2 n/256}$ (as $|S| \geq n/(8\ell_0) = \log^2 n/8$), which is summable over $\ell < \ell_0$ to $o(1/n)$. The identity behind (4) holds for the average over $c \in S$ as well as over all c , so with $\theta_0 = \frac{1}{8}$, $\sum_{\ell < \ell_0} \mathbb{P}[B, d_{12} = \ell, |C_1| \geq n/2 + \ell, \theta_S \geq \theta_0] \leq 36\ell_0/((n-2)\theta_0) = O(1/\log^2 n)$. The case $d_{21} = \ell$ is symmetric. Altogether $\mathbb{P}[\text{Flip} = \emptyset] = o(1/\log n)$. $\square$

What the hypothesis says, and what it costs. A ladder pair enters U when one of its T candidate column pairs has a short column cycle through x_k that is clean — a configuration of $O(\log^2 n)$ cells ($4t$ cells of rows x_k, y_k to locate the candidate, and the cycle itself). Heuristically a candidate's column cycle has length $\leq \ell_1$ with probability $\approx \ell_1/n$ (the data of §(d) show the lengths spread like a uniform variable on $\{2, \dots, n\}$), is clean with probability $\approx e^{-2s\ell_1/n}$ for a family of s pairs, and the chosen pair avoids the $\approx 2Ts$ candidate columns of the other pairs with probability $\approx (1 - 2Ts/n)^2$, so

$$\mathbb{E}|U| \approx s \cdot \frac{T\ell_1}{n} e^{-2s\ell_1/n} \left(1 - \frac{2Ts}{n}\right)^2 \asymp \log^2 n \quad \text{for } s = \frac{n}{\log^2 n}, T = \frac{\log^2 n}{4}, \ell_1 = \frac{\log^2 n}{2},$$

against the $C \log \log n$ ($C > 1/\log 2$) that (i) needs; the constant is small ($\approx 1/80$), so the heuristic excess over the threshold is a large- n statement. Compared with the fixed matching of (c), which offers $\approx n/2$ coordinates per pair of which a constant number are short and clean and a fraction ≈ 0.19 of those separated, the adaptive rule offers only $T = O(\log^2 n)$ coordinates per pair, all separated: the genericity constant is gone, and in exchange membership in U is a rare event per pair, so (i) is a lower-tail statement about rare local events at rows selected by the square — of the same shape as a statement that some ladder pair has a short ρ_{x_k, y_k} -cycle through p avoiding p' (which would make it flippable outright), now with a column cycle of a free pair through x_k in its place. There is a second regime: with $s \approx T \approx \ell_1 \approx \sqrt{n}$ a pair enters U with probability bounded below and $\mathbb{E}|U| \asymp \sqrt{n}$, so (i) becomes a second-moment statement about $\sqrt{n}$ events of constant probability, at the price of the arcs of ρ_k at p, p' exceeding $\sqrt{n}$ for a constant fraction of the pairs. In both regimes the lengths of the arcs of ρ_k at p, p' — which the fixed-matching version needed to be macroscopic for a positive orbit weight — enter only through the number of candidates $\min(T, \mu_k - 1)$. The collision rule in U (the chosen pair must avoid the candidate columns of *all* other pairs) is what makes U orbit-invariant — a turn can shrink the candidate set of an excluded pair, so the obvious greedy refinement is not invariant — and it is crude: it is why the asymptotic regime $2Ts/n \leq \frac{1}{2}$ is out of reach of sampling at $n \leq 100$.

(d) Data. All scripts and logs are in the repository [6], under `s13*.py` and `runs/s13/orbit/`. *Adaptive coordinates.* On true-marked instances at $n = 30, 50, 100$ (family the first $\min(K_0 - 1, n/8)$ ladder pairs, $T = 6, \ell_1 = 6, 8, 10$; 2400, 3000, 900 instances), for a random $k \in U$ the turn of Z_k was applied and checked to toggle the status of (x_k, y_k) , to leave the status, $\mu_i, t_0(l), C_l$ of every other pair and the set U unchanged, and to preserve the type of rows 1, 2 and the frame (682, 697, 215 turns, no failure; 444 and 160 more with families of 12 and 30 pairs); the bound $\mathbb{P}[\text{Flip}_I = \emptyset] \leq \mathbb{E}[2^{-|U|}]$ holds in every K_0 class. At these sizes 30%, 23%, 16% of the ladder pairs have a good candidate and 13%, 5%, 3% are in U (8% and 1.4% with the family of 30 pairs at $n = 100$): $2Ts \geq n$ there, so the collision rule removes most pairs. *Column-cycle lengths and the fixed-matching coordinates.* The length of the column cycle of a free pair through a ladder row is spread like a uniform variable on $\{2, \dots, n\}$ (about $1/n$ per length; the turn is legal for 98% of the 2-cycles and 64% of the long cycles, and avoids y for 97% resp. 27% of them, at $n = 100$). For the fixed matching of (c): $\mathbb{P}[(q, c) \text{ separated} \mid \text{clean, length } \ell] = 0.193\text{--}0.201$ for $\ell = 2, \dots, 8$ against 0.195 over all column pairs ($n = 100$), i.e. no detectable dependence on the cycle length; with $|\mathcal{T}| \approx 42$ coordinates the orbit average of the parallel bit is 0.478 from a crossed and 0.523 from a parallel start ($n = 100$); with the fixed matching of consecutive free columns $|\mathcal{T}| = 1.7$ resp. 3.0 clean coordinates per pair ($n = 50, 100$) and $\mathbb{E}[\bar{y}] = 0.159$ resp.

0.189; and the orbit weights of different ladder pairs are uncorrelated to within sampling error ($\text{Var}(Y_I)/(|I|\text{Var } \bar{y}) = 0.99$ and 1.14 at $n = 50, 100$, the excess coming from the variation of $|I|$). The ladder data of Remark 4.4 complete the picture: $\mathbb{P}[\text{Flip} = \emptyset] \approx 4/n$, as if the ladder coins were fair and independent.

6 What remains

Hypothesis 5.6 is a statement about short cycles at the frame-selected rows: that enough ladder pairs (x_k, y_k) have, at one of the prescribed column pairs $(\rho_k^t(p), \rho_k^t(p'))$, a column cycle through x_k of length $O(\log^2 n)$ that avoids the other ladder rows and meets $\{1, 2\}$ in 0 or 2 rows; and that the arcs of ρ_k at p, p' are not shorter than T for most k . Both are local (polylogarithmic) configurations whose heuristic frequency is, for large n , far above what is needed; the difficulty is entirely that the rows are selected by the frame of the mark and the whole space is conditioned on the type of rows 1, 2. We record what the available tools do and do not give.

Unconditional facts. For two generic rows of a uniform square, Proposition 3.2(i) gives, by Markov's inequality on the number of columns on short cycles, $\mathbb{P}[\text{all cycles of } \rho_{x,y} \text{ have length } \leq \varepsilon n] \leq 2\varepsilon/(1 - \varepsilon)$; so a generic pair has a cycle longer than $n/5$ with probability $\geq \frac{1}{2}$, and the column-cycle analogue holds by transposition. Kwan and Sudakov's bound $\mathbb{P}[N_2 \geq t] = e^{-\Omega(t \log t)}$ on the number of 2-cycles of two fixed rows [4, Thm. 3] is proved by joining switchings of the same kind as those of Proposition 3.2 and is likewise an upper bound on short cycles. The number of completions of any $2 \times n$ rectangle is within $e^{O(n \log^2 n)}$ of any other (van der Waerden and Brégman; [4, Prop. 5]), so any property of uniform squares with failure probability $e^{-\omega(n \log^2 n)}$ — such as the intercalate concentration of Kwan, Sah and Sawhney [5] — holds conditionally on any fixed rows 1, 2 and mark, hence in X for every λ, α, β ; but the natural failure probability of a statement about $m \leq n$ prescribed row pairs is $e^{-\Omega(m)}$, which this transfer does not reach.

Why the fixed-matching version is harder. With a fixed matching the orbit weight $\bar{y}_k$ of Proposition 5.3 needs a lower bound on the probability that a generic column pair is interleaved with $\{p, p'\}$ in ρ_{x_k, y_k} — an instance of the statement “two given pairs of lines of a random Latin square are interleaved in the cycle structure of a third pair with probability in $[c, 1 - c]$ ” inside X . Without the conditioning its upper half is elementary and its lower half reduces to macroscopic cycles (by conjugation invariance the interleaving probability given the cycle type of $\rho_{x,y}$ is $\frac{1}{3} \sum_i (m_i)_4 / (n)_4 + 2 \sum_{i \neq j} (m_i)_2 (m_j)_2 / (n)_4 \leq \frac{1}{3} + O(1/n)$, and $\geq \frac{1}{3} \mathbb{P}[W \mid \text{type}]^3 - O(1/n)$ by Jensen, W the event that two given columns share a cycle). Inside X the exact tools are the free trades of rows other than 1, 2 (which preserve interleaving) and the legal column turns (which are legal exactly off the interleaved configurations), and every identity they produce is fair: it equates two expectations with random weights whose lower bounds are again of the same form. Lower bounds on “different cycles” events are, in the language of [1], of the type of the direction of their Lemma 3.12 whose proof has the gap: they require a repair of the same-cycle configurations with a rigid inverse, and the repairs available in X (trades or cross-switches of a free pair of third rows) have unbounded multiplicity. The one case in which we found a rigid inverse is CGW's own case 3 with $\min(\alpha, \beta) = 2$, which gives $\mathbb{P}_X[A] \geq \frac{1}{4}$ for marks at distance 2 (Remark 3.4); it does not extend to other distances for a reason of type rather than multiplicity. The adaptive coordinates sidestep the question: the coordinate is placed where it is separated by construction, and the price is paid in rarity rather than in a constant we cannot prove.

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